

TrES-1b

R-2558

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_260722 · created 2026-08-26T22:55:56+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461243.7584 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.053
Transit depth (Rp/Rs)^2	3.36 +/- 1.93 [%]
Orbital Inclination (inc)	85.57 +/- 1.47
Airmass coefficient 1 (a1)	29.14 +/- 16.34
Airmass coefficient 2 (a2)	-2.97 +/- 1.47
Scatter in the residuals of the lightcurve fit is	53.81 %
Best Comparison Star	#2 - [594, 217]
Optimal Aperture	1.58
Optimal Annulus	7.0
Transit Duration (day)	0.063 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.183 ± 0.053 vs 0.1358 (deviation 0.89σ)
Duration fit vs published	0.063 d vs 0.1042 d (deviation 1.72σ)
Mid-time O–C	-6.2 min
Depth SNR	3.5
Residual scatter	53.81 %

Light curve

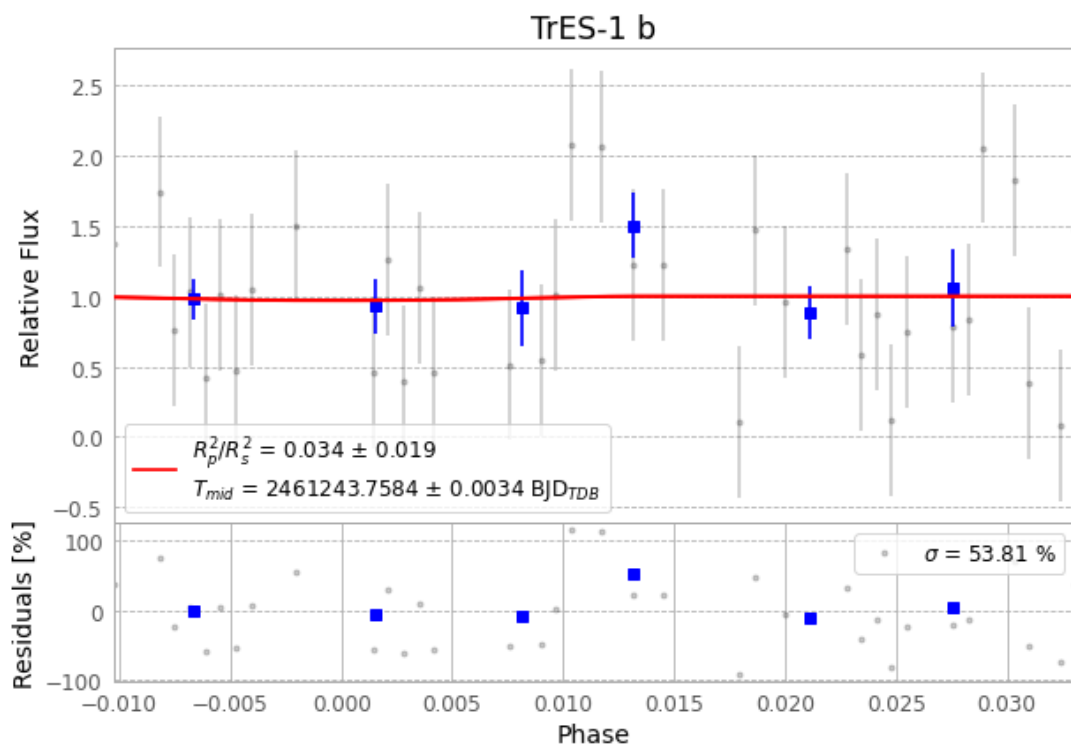


Figure 1 — FinalLightCurve_TrES-1 b_2026-07-21.png (SHA-256 5bf80971c8620fdc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [585, 258], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 eae92bb79773e1d1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), TRES-1260722051515.FITS → TRES-1260722083015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-260722.log (3dbc28c80b80...), FinalLightCurve_TrES-1 b_2026-07-21.png (5bf80971c862...), FinalLightCurve_TrES-1 b_2026-07-21.csv (d4eb523bc519...)

Compute resources

Wall time 138.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 22:55Z.